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Mapping AI Literacy in Design Thinking: Where and How AI Tools Impact Learning Processes

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Introduction

The rapid growth of artificial intelligence (AI) in education has raised questions about how AI tools shape student learning in design thinking contexts. This scoping review synthesizes empirical studies that integrate AI tools into design thinking learning activities and report AI literacy outcomes. Following Arksey and O'Malley's (2005) framework, searches across EBSCO, Scopus, Web of Science, and IEEE Xplore yielded 18 empirical studies (2020–2025). Most studies focused on K–12 and higher education learners, with publication activity concentrated after 2022 and peaking in 2025. Using deductive content analysis coding, we mapped each study to five design thinking processes (Empathize, Define, Ideate, Prototype, Test) and four AI literacy dimensions (Know and Understand AI; Use and Apply AI; Evaluate and Create AI; AI Ethics). Findings show an asymmetric pattern: studies cluster in Ideate, Prototype, and Test, while Empathize is least examined. Across studies, Know & Understand AI is most emphasized, followed by Use & Apply AI; Evaluate and Create AI and AI Ethics are less consistently addressed and assessed. Future research should strengthen phase-level alignment, broaden learner and disciplinary representation, and adopt clearer frameworks and validated measures to improve comparability and cumulative knowledge.

Findings

		Design Thinking					Influence Direction	AI Literacy			
Number	Study	Emphathize	Define	Ideate	Protoype	Test		Know & Understand AI	Use & Apply AI	Evaluate & Create AI	AI Ethics
1	(Solyst et al., 2023)						→				
2	(Dwivedi et al., 2024)						→				
3	(Dipaola et al., 2023)						→				
4	(Han & Han, 2025)						↔				
5	(Akman, 2025)						→				
6	(Smith et al., 2025)						→				
7	(Wang et al., 2025)						↔				
8	(Kim & Lee, 2023)						←				
9	(Marçal et al., 2025)						↔				
10	(Long et al., 2023)						→				
11	(Yue et al., 2025)						→				
12	(Imjai et al., 2024)						↔				
13	(Rana et al., 2025)						↔				
14	(Yavuz et al., 2025)						→				
15	(Rong et al., 2024)						←				
16	(Kim et al., 2025)						↔				
17	(Cheng et al., n.d.)						→				
18	(Hsu & Hsu, 2024)						↔				

Table 1. Mapping of Design Thinking Processes, AI Literacy Dimensions, and Direction of Influence Across Included Studies

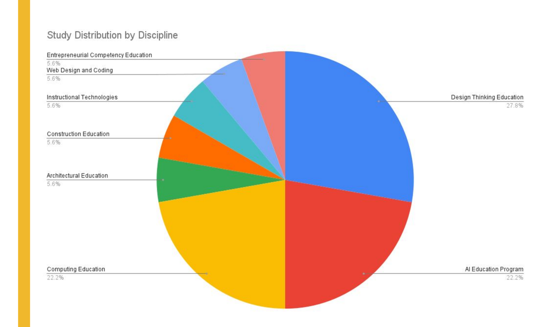

Figure 1. Study Distribution by Discipline

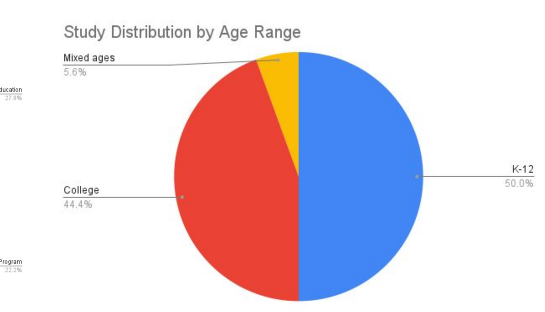

Figure 2. Study Distribution by Age Range

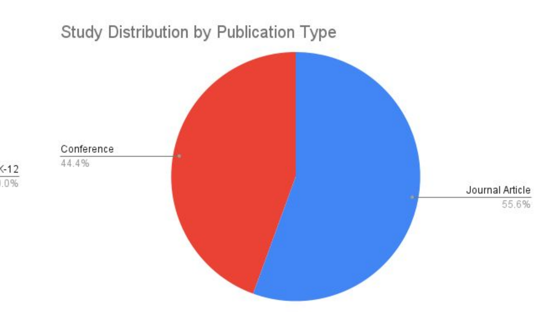

Figure 3. Study Distribution by Publication Type

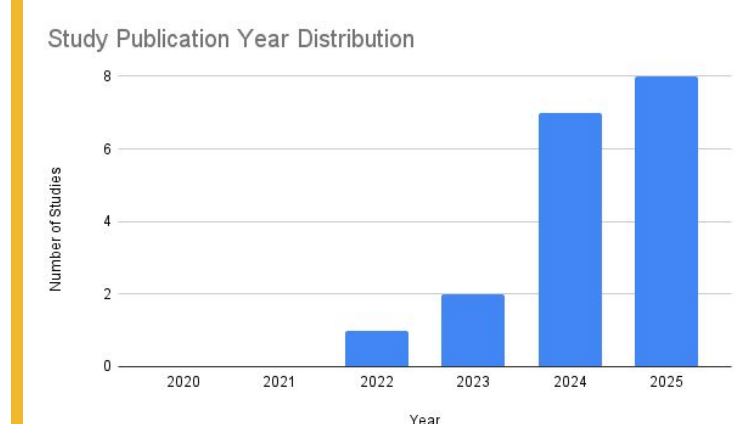

Figure 4. Study Publication Year Distribution

Figure 5. Study Geographic Distribution

Purpose & Research Question(s)

Purpose

This scoping review maps how AI tools are being integrated into design thinking learning activities from 2020–2025. The study also codes for the different AI literacy dimensions emphasized in each study. The goal is to identify where AI tools have the strongest learning impact and where evidence gaps remain.

Research Questions

RQ1: What is the current scope and nature of academic literature from the past five years on AI literacy within the context of Design Thinking?

RQ2: How can current conceptualizations of AI literacy be mapped onto the design thinking process to identify where AI tools have the most impact?

Methods

Framework by Arksey & O'Malley (2005)

Stage 1 (Identifying the Research Questions)

Defined questions on AI literacy in design thinking and where artificial intelligence tools impact the design thinking process.

Stage 2 (Identifying Relevant Studies)

Searched EBSCO, Scopus, Web of Science, and IEEE Xplore (2020–2025; English; peer-reviewed).

Stage 3 (Study Selection)

Screened titles/abstracts and full texts; included 18 empirical studies linking AI tools, design thinking, and AI literacy.

Stage 4 (Charting the Data)

Extracted study characteristics and coded design thinking phases (Empathize, Define, Ideate, Prototype, Test) and AI literacy dimensions (Know and Understand AI; Use and Apply Artificial Intelligence; Evaluate and Create AI; AI Ethics).

Stage 5 (Collating, Summarizing, & Reporting)

Summarized trends & mapped design thinking phases to AI literacy dimensions to identify impact areas and gaps using a content analysis approach.

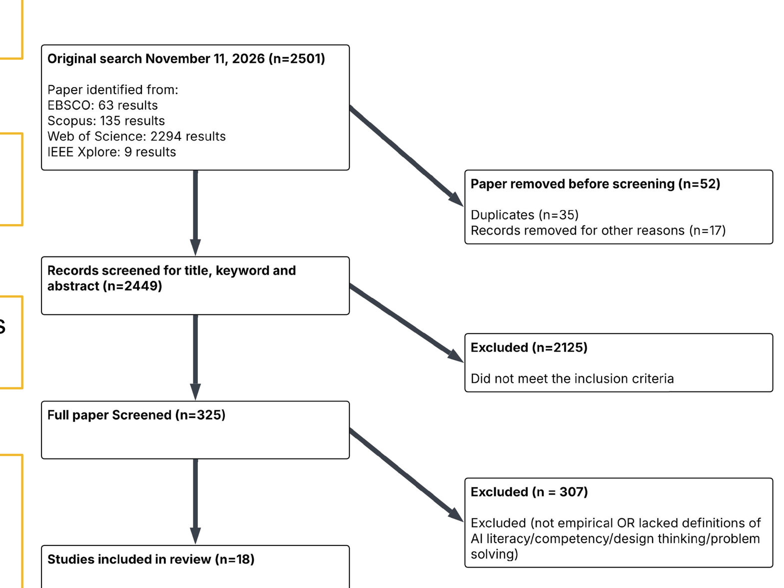

Figure 6. Study selection

Discussion and Conclusion

Discussion

- Post-2022 GenAI drives a rapidly growing literature (Figure 4).
- The evidence base is concentrated in a few disciplines and regions, suggesting current conclusions may not generalize well across broader design fields or global contexts (Figure 1-3, 5).
- Inconsistent AI literacy definitions/measures limit comparability and call for shared frameworks and validated tools.

Conclusion

- Design thinking and AI literacy are co-developed. AI tools most strongly shape learning in later design thinking phases, not early empathy work. Empathize and Define phases of design thinking remain understudied (Table 1).
- Next steps: study Empathize/Define, standardize AI literacy frameworks, and use validated assessments.